

7. Integration Techniques

7.8. Improper Integrals

7.8b Learning Objectives

- Identify and rewrite type 2 improper integrals as limits of proper integrals
- Use the comparison test to determine whether an integral converges or diverges

7.8.3. Type 2 Improper Integrals

Definition 7.8.1 Type 2 Improper Integrals

Improper integrals of Type 2 are definite integrals with integrands that are *not continuous* within the limits of integration.

Suppose f is a function.

1. If f is continuous on $(a, b]$, but not continuous at a , then

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow a^+} \int_t^b f(x) \, dx.$$

2. If f is continuous on $(a, b]$, but not continuous at b , then

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow b^-} \int_a^t f(x) \, dx.$$

3. If f is continuous on $[a, c) \cup (c, b]$, but not continuous at c in (a, b) , then

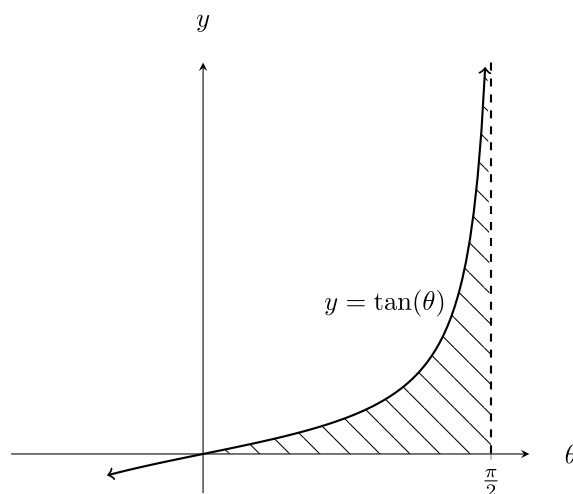
$$\begin{aligned} \int_a^b f(x) \, dx &= \int_a^c f(x) \, dx + \int_c^b f(x) \, dx \\ &= \lim_{t \rightarrow c^-} \int_a^t f(x) \, dx + \lim_{r \rightarrow c^+} \int_r^b f(x) \, dx \end{aligned}$$

If the limit is finite, then we say that the definite integral **converges** and the integral is the **value** of the limit. If the limit is not finite, we say the definite integral **diverges**.

Example 7.8.1

Does $\int_0^{\frac{\pi}{2}} \tan(\theta) \, d\theta$ converge? If so, what is its value?

 Solution



Notice, this type of improper integral involves the cases where the function has a vertical asymptote, so there may be infinite or finite area. It depends on how “quickly” the curve approaches the asymptote.

The upper bound on the integral is where the discontinuity is. So we must take the limit as the upper bound approaches $\frac{\pi}{2}$ from below. Specifically,

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \tan(\theta) \, d\theta &= \lim_{t \rightarrow (\frac{\pi}{2})^-} \int_0^t \tan(\theta) \, d\theta \\ &= \lim_{t \rightarrow (\frac{\pi}{2})^-} \int_0^t \frac{\sin(\theta)}{\cos(\theta)} \, d\theta\end{aligned}$$

Let $u = \cos(\theta)$, $du = -\sin(\theta) \, d\theta$, $u(0) = 1$, $u(t) = \cos(t)$ then

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \tan(\theta) \, d\theta &= \lim_{t \rightarrow (\frac{\pi}{2})^-} \int_1^{\cos(t)} \frac{\sin(\theta)}{\cos(\theta)} \, d\theta \\ &= \lim_{t \rightarrow (\frac{\pi}{2})^-} (-1) \int_1^{\cos(t)} \frac{1}{u} \, du \\ &= \lim_{t \rightarrow (\frac{\pi}{2})^-} (-1) \ln|u| \Big|_1^{\cos(t)} \\ &= \lim_{t \rightarrow (\frac{\pi}{2})^-} -\ln|\sec(t)| \\ &= \infty\end{aligned}$$

because this limit diverges, the integral diverges.

Note, this implies the shaded area is not finite.

Example 7.8.2

Evaluate $\int_0^3 \frac{1}{(s-1)^{\frac{2}{3}}} \, ds$ if possible, or show it diverges.

Note

This function is undefined at $s = 1$, which is in the interval we are integrating over. If we were to just integrate this without taking care of the vertical asymptote, we'd possibly get a value that is not correct. Again, the hardest part of these integrals is the notation and treating it with care using limits.

Solution

Because the function is discontinuous at $s = 1$,

$$\begin{aligned}\int_0^3 \frac{1}{(s-1)^{\frac{2}{3}}} ds &= \lim_{t \rightarrow 1^-} \int_0^t \frac{1}{(s-1)^{\frac{2}{3}}} ds + \lim_{r \rightarrow 1^+} \int_r^3 \frac{1}{(s-1)^{\frac{2}{3}}} ds \\&= \lim_{t \rightarrow 1^-} \left[3(s-1)^{\frac{1}{3}} \right]_0^t + \lim_{r \rightarrow 1^+} \left[3(s-1)^{\frac{1}{3}} \right]_r^3 \\&= \lim_{t \rightarrow 1^-} \left[3(t-1)^{\frac{1}{3}} - 3(-1) \right] + \lim_{r \rightarrow 1^+} \left[3(2)^{\frac{1}{3}} - 3(r-1)^{\frac{1}{3}} \right] \\&= 3 + 3(2)^{\frac{1}{3}}\end{aligned}$$

7.8.4. Direct Comparison Test for Improper Integrals

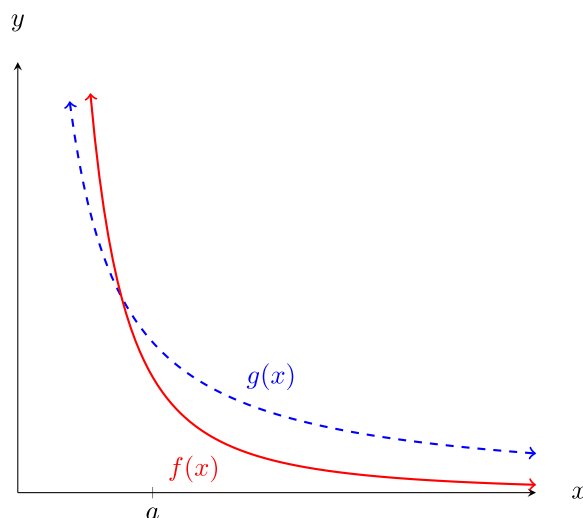
Theorem 7.8.1 Direct Comparison Test for Improper Integrals

Let f and g be continuous on $[a, \infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \geq a$. Then,

1. If $\int_a^\infty g(x) dx$ converges, then we can conclude that $\int_a^\infty f(x) dx$ will converge also.
2. Alternatively, if $\int_a^\infty f(x) dx$ diverges, then we can conclude that $\int_a^\infty g(x) dx$ will diverge also.

Note

Why does this make sense?



In the first case, if we know that the larger of the 2 areas is finite (i.e. that $\int_a^\infty g(x) dx$ converges), then we know that the area under f is less (i.e. we can conclude that $\int_a^\infty f(x) dx$ must also converge).

In the second case, we instead know the area under f is not finite (i.e. $\int_a^\infty f(x) dx$ diverges). Then, because g is always above f , the area under g is greater than the area under f , which is not finite. Thus, the area under g cannot be finite (i.e. $\int_a^\infty g(x) dx$ diverges).

Example 7.8.3

Does $\int_0^\infty e^{-x^2} dx$ converge?

Note

This integral is still one we cannot do analytically (without using approximations). That is to say that the antiderivative of e^{-x^2} is not one we can write down. So, to answer the question we will use the comparison test to compare this integral to one we know more about.

Coming up with the function we compare to is an art, and because we are only human we'll keep it pretty simple in this class. (Large language models and people with too much time on their hands can come up with incredibly obtuse and non-obvious functions to use the comparison test on, and when people do it it can be impressive)

Solution

Observe e^{-x^2} is kind of like e^{-x} . Note, $e^{-x} > e^{-x^2}$ for $x > 1$ and, importantly, we can integrate e^{-x} .

First, we show $e^{-x} > e^{-x^2}$ for $x > 1$ (yes this is part of a complete solution, and you would have to do something similar with your comparison functions): For $x > 1$ we have

$$\begin{aligned} x^2 &> x \\ \text{so } -x^2 &< -x \\ \text{so } e^{-x^2} &< e^{-x}. \end{aligned}$$

Next, we show $\int_1^\infty e^{-x} dx$ converges:

$$\begin{aligned} \int_1^\infty e^{-x} dx &= \lim_{t \rightarrow \infty} \int_1^t e^{-x} dx \\ &= \lim_{t \rightarrow \infty} -e^{-x} \Big|_1^t \\ &= \lim_{t \rightarrow \infty} e^{-1} - 1(e^t) \\ &= e^{-1} - 0 = e^{-1} \end{aligned}$$

Because e^{-1} is finite $\int_1^\infty e^{-x} dx$ converges.

Lastly, we need to connect this to $\int_0^\infty e^{-x^2} dx$: If it converges,

$$\int_0^{\infty} e^{-x^2} dx = \int_0^1 e^{-x^2} dx + \int_1^{\infty} e^{-x^2} dx$$

By the direct comparison test, since $\int_1^{\infty} e^{-x} dx$ converges and $e^{-x} > e^{-x^2}$ for $x > 1$ then $\int_1^{\infty} e^{-x^2} dx$ must also converge. Finally, since $\int_0^1 e^{-x^2} dx$ is a finite number (finite continuous function integrated over a finite interval) the sum

$$\int_0^{\infty} e^{-x^2} dx = \int_0^1 e^{-x^2} dx + \int_1^{\infty} e^{-x^2} dx$$

must also converge.

Note: We don't know what the integral will converge to still! Just that the improper definite integral is a finite number.

Note

This solution goes into a little extra detail than we would usually require, specifically the breaking the integral up from 0 to 1 then 1 to ∞ . This is because our direct comparison test had a be both the starting point of the integral and the point where $g(x) \geq f(x)$ from that point onward. This is not required. So long as one function is larger than the other *eventually* (and forever after that point) direct comparison test will work.

7.8b Section Summary:

- We discussed type 2 improper integrals, those with discontinuities on the interval we integrate over.
- We discussed when improper integrals of type 2 converge or diverge.
- We used direct comparison test to conclude if an integral we cannot do converges or diverges.